

5(a)(i)	<u>Morbillivirus</u> ;	1
5(a)(ii)	any three from: 1 (pathogen is transmitted) in airborne droplets / as an aerosol ; A droplet infection / aerosol infection 2 breathed / sneezed / coughed / AW, out by infected person ; 3 (droplets) breathed in by, healthy / uninfected, person ; 4 touching surface with virus and, breathing into lungs / puts fingers in, nose / mouth ;	3
5(b)	any three from: 1 increase and decrease / fluctuates / AW ; A between 2015 and 2019, over some of the years or just one year A increase, decrease and then increase for any one year 2015 to 2018 2 use of data to support mp1 ; 3 numbers for January 2019 are, highest / higher than other years or numbers for March 2019 are, lowest / lower than other years ; 4 use of data to support mp3 ;	3

Question	Answer	Marks
5(c)	<i>ignore</i> mutation and vaccine evasion / vaccine resistance <i>ignore refs to cost</i> any four from: <i>transmission and susceptibility (from mp1 to mp7)</i> 1 infected person can infect many people / spreads easily in crowded conditions ; 2 measles spreads to children before they are vaccinated ; 3 vaccine cannot be given soon after birth because of passive immunity ; 4 immunisation rates do not reach 100% / ref to difficulty in reaching herd immunity / herd immunity is not achieved ; 5 herd immunity needed for protection of, unvaccinated people / babies ; 6 some people have, no / poor, response to vaccine ; A vaccine not always effective 7 <i>ref. to</i> malnutrition / <i>ref. to</i> lack of vitamin A ; <i>reasons for low immunisation rates (from mp8 to mp12)</i> 8 difficult to reach all people, with example ; e.g. in rural areas areas of poor housing in big cities 9 collapse of vaccination programmes as a result of war / AW ; 10 lack of, trained professionals / health facilities ; 11 many children do not receive boosters ; 12 reluctance to be vaccinated / <i>ref. to</i> anti-vaccination campaigns / ref to MMR or triple vaccine / lack of awareness about beneficial effects of vaccine / <i>ref. to</i> cultural or religious reasons / AW ; 13 AVP ; (<i>for either section of the answer</i>) e.g. <i>ref. to</i> any specific difficulty with contact tracing <i>ref. to</i> thermostability of vaccine and maintaining a cold chain migrants, with measles / from an area with an outbreak of measles	4